Reg. No. | | | | | | | | | | |

[18MSCH1H04]

FIRST SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER 2019

SPECTROSCOPY - I

Time :03 Hours **Maximum Marks:70**

Instructions: 1. **Question 1 is compulsory, carries 10 marks.**

2. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. Answer any FIVE of the following questions: **5×2=10**

a. Arrange the different types of electronic transitions in the increasing order of energy with a neat sketch.

b. How to differentiate *ortho*, *meta* and *para* substitution in an aromatic ring using IR spectroscopy?

c. Explain the fundamental frequencies of nitrile and isocyanate groups.

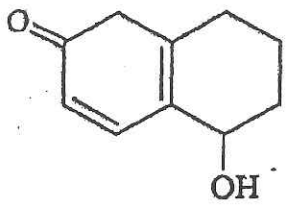
d. Which of the following molecules would show rotational spectrum: Br_2, HBr and CS_2? Why?

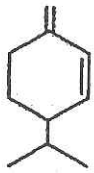
e. Define chemical shift in NMR spectroscopy. How is it calculated?

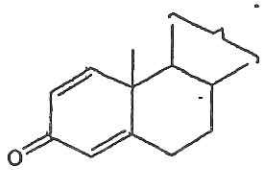
f. Explain the effect of hybridization in ^{13}C NMR spectroscopy.

2. a. Calculate the λ_{max} value for the following conjugated dienes and trienes in ethanolic solution by using Woodward-Fieser rules.

i.

ii. 

iii. 

iv. 

(2.5 × 4 = 10)

b. Explain in detail any two applications of UV-Vis spectrophotometer. **05**

3. a. How to analyze an IR spectrum? Explain in detail. **10**

b. What are the differences between IR and Raman spectroscopy? **05**

4. a. How do you estimate the drug molecule's concentration in the pharmaceutical industry by using UV Spectroscopy? **10**

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b. Explain the fundamental frequencies of nitro compounds, carboxylate salts a. amino acids with one example each. **05**

5. a. The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the rotational constant (B) and bond length of CO. The relative atomic weight C =12.00 and O = 15.9994, the absolute mass of H=1.67343×10^{-27} kg. **10**

b. Describe instrumentation of microwave spectroscopy. **05**

6. a. Discuss the origin of NMR peak splitting in detail. Explain the splitting pattern and peak intensity of ethyl group (CH_3CH_2-). **10**

b. What is magnetic anisotropy in NMR spectroscopy? Explain with an example. **05**

7. a. What is simplification of complex NMR spectrum? Discuss the various methods of simplifying complex NMR spectrum. **10**

b. Explain COSY and HETCOR techniques with an example each. **05**